Deterministic Optimization

SOCP, SDP Examples

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Experiment Design

SOCP, SDP Examples

Learning Objectives for this lesson

- For this lesson, we focus on experiment design
- Discover an SDP relaxation of an experiment design problem

Estimation Problem

- Consider a problem of estimating vector $x \in R^n$ from measurements or experiments

$$y_i = a_i^\top x + w_i,\ i = 1,2,\ldots,m$$

where w_i's are measurement noise, Gaussian with zero mean and unit variance

- The maximum likelihood estimate of x is given by
  - $\hat{x} = \left(\sum_{i=1}^{m} a_i a_i^\top\right)^{-1} \sum_{i=1}^{m} y_i a_i$
- Estimation error $e = \hat{x} - x$ has zero mean and covariance matrix
  - $E = E[ee^\top] = \left(\sum_{i=1}^{m} a_i a_i^\top\right)^{-1}$

The Goal of Experiment Design

- Suppose the vectors $a_1, \ldots, a_m$ can be chosen among p possible test vectors $v_1, \ldots, v_p \in R^n$

- The goal of *Experiment Design* is to choose the vectors a_i from among the above choices, to minimize error covariance E in a certain sense, i.e. we want to design a set of measurement (an experiment) to get the most informative estimate of x

A Basic Formulation of Experiment Design

- Let z_j be the number of experiments for which a_i is chosen to have the value v_j
  - $z_1 + z_2 + \cdots + z_p = m$, $z_i \geq 0$, integer
- Error covariance is
  - $E = \left(\sum_{i=1}^{m} a_i a_i^\top\right)^{-1} = \left(\sum_{j=1}^{p} z_j v_j v_j^\top\right)^{-1}$
- So a basic formulation of experiment design is given as
  - $\min\left(\sum_{j=1}^{p} z_j v_j v_j^\top\right)^{-1}$
  - S.t. $z_1 + z_2 + \cdots + z_p = m$
  - $z_i \geq 0$, integer, $i = 1, \ldots, p$

Multi-objective
Nonlinear integer program
Very challenging!

A Practical Formulation

- Multi-objective → a single objective of minimize the trace of E
  - $tr(E) = tr(E[ee^\top]) = E[tr(ee^\top)] = E[||e||_2^2] = \text{mean of error squared}$
- Integer variable → continuous relaxation
- $\min \; tr\left(\sum_{j=1}^{p} z_j v_j v_j^\top\right)^{-1}$
- s.t. $z_1 + z_2 + \cdots + z_p = m$, $z_i \geq 0$, $\forall i = 1, \ldots, p$
- Which is equivalent to
- $\min \frac{1}{m} tr\left(\sum_{j=1}^{p} z_j v_j v_j^\top\right)^{-1}$
- s.t. $z_1 + z_2 + \cdots + z_p = 1$, $z_i \geq 0$, $\forall i = 1, \ldots, p$

SDP Reformulation

- $\min \frac{1}{m} tr\left(\sum_{j=1}^{p} z_j v_j v_j^\top\right)^{-1}$
- s.t. $z_1 + z_2 + \cdots + z_p = 1,\ z_j \geq 0,\ \forall j = 1, \ldots, p$ (1)

Is equivalent to

- $\min tr\left(\sum_{j=1}^{p} z_j v_j v_j^\top\right)^{-1}$
- s.t. (1)
- $\min \sum_i e_i^\top \left(\sum_{j=1}^{p} z_j v_j v_j^\top\right)^{-1} e_i$ ➔
- $\min \sum_i u_i \ \ s.t.\, e_i^\top \left(\sum_{j=1}^{p} z_j v_j v_j^\top\right)^{-1} e_i \leq u_i\ \forall i = 1, \ldots, n,$ (1)

SDP Reformulation cont.

- $e_i^\top \left(\sum_{j=1}^p z_j v_j v_j^\top\right)^{-1} e_i \leq u_i$ can be rewritten as
- $\begin{bmatrix} u_i & e_i^\top \\ e_i & \sum_{j=1}^p z_j v_j v_j^\top \end{bmatrix} \succcurlyeq 0 \;\; \forall i = 1, \ldots n$ [Schur Complement]
- Put everything together, we get an SDP reformulation
    - $\min \sum_i u_i$
    - S.t. $\begin{bmatrix} u_i & e_i^\top \\ e_i & \sum_{j=1}^p z_j v_j v_j^\top \end{bmatrix} \succcurlyeq 0 \;\; \forall i = 1, \ldots n$
    - $z_1 + z_2 + \cdots + z_p = 1,\ z_j \geq 0,\ \forall j = 1, \ldots, p$

Summary

- We learned:
  - What is the experiment design problem
  - How to formulate an SDP relaxation